

EEN1095 Git Repository and Development Record

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Project Title	Design and Evaluation of Low-Sensitivity Digital Filters Implemented in Python Under Varying Numerical Precision Levels	Supervisor	Martin Collier
Implementation Period	Week 1 - Week 2	Version Date	22/05/2026
Link to the Repository	Not yet created	Shared to Supervisor	No

Guidance: Maintain this as one cumulative document throughout EEN1095. Update it every two weeks and do not remove earlier records. Use it to summarise sustained implementation and development work evidenced through your Git repository or equivalent version-controlled development record. Each entry should summarise the development work completed, the principal files or components affected, and the key evidence of progress. Where applicable, relevant screenshots of the Git commit history should also be provided.

Development Record:

Weeks 1-2	
Main implementation / development work completed	Python multi-precision evaluation framework development. Set up Python 3.11 environment with NumPy, SciPy, and mpmath. Implemented filter_design.py for IIR and FIR filter design using SciPy. Implemented metrics.py defining all five performance metrics. Implemented experiment.py as the main experimental runner. Validated precision control for float16, float32, float64 via NumPy dtype and mpmath via mp.dps=50. Ran pilot experiments for Direct-Form IIR filters at orders 10 and 20 across all four precision levels.
Key files, folders, or components added / updated	filter_design.py — Core filter design and precision conversion module: design_iir_filter(), design_fir_filter(), convert_to_precision(), get_poles_from_a(), compute_freq_response(). metrics.py — Five performance metrics module: pole_displacement(), stability_margin(), sensitivity_norm(), freq_response_deviation(), roundoff_noise(). experiment.py — Main experimental evaluation runner:

	run_iir_experiment(), run_fir_experiment(), run_all_experiments(). results.json — Pilot experimental results dataset for Direct-Form IIR configurations across all precision levels.
Evidence of versioning / commits / iterations	Git repository not yet created — code files are locally version tracked. Initial commit planned once repository is set up in Weeks 3-4. Current iteration: v0.1 — framework setup and pilot testing complete. Three main Python modules implemented and validated: filter_design.py, metrics.py, experiment.py. Pilot results for Direct-Form IIR filters at orders 10 and 20 stored in results.json.
Testing, debugging, or validation carried out	Precision control validation: confirmed float16, float32, float64 dtype selection via NumPy astype() correctly reduces coefficient precision. mpmath mp.dps=50 confirmed providing 50 decimal digit precision. Filter design validated against known analytical frequency responses. Metrics validated on order-10 Butterworth filter. Key finding confirmed: Direct-Form order-20 filters become UNSTABLE at float16 — poles breach the unit circle. All FIR filters remain stable at all precision levels.
Problems encountered and how they were addressed	Minor issue: mpmath coefficient conversion required handling via mpmath.mpf(str(c)) to preserve exact decimal precision during conversion from SciPy float64. Resolved successfully. Minor issue: NumPy overflow warnings during float16 sensitivity norm computation. Resolved by wrapping in try-except and filtering expected overflow warnings.
Next development steps	Weeks 3-4: Create Git repository and push all current framework files. Implement Ladder filter Python module based on Bruton 1975 [1] using cascaded lossless two-port sections. Begin Wave Digital Filter implementation based on Fettweis 1986 [2] and Yli-Kaakinen 2006 [9]. Validate Ladder and WDF implementations against Direct-Form baseline at float64. Run full order-30 experiments for all four IIR filter types across all precision levels.

Weeks 3-4	
Main implementation / development work completed	
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	Enter a short summary of commit activity here. Add commit-history screenshots if applicable.
Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	
Next development steps	Outline the planned development tasks for the next two-week period.

Weeks 5-6	
Main implementation / development work completed	
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	Enter a short summary of commit activity here. Add commit-history screenshots if applicable.
Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	
Next development steps	Outline the planned development tasks for the next two-week period.

Weeks 7-8	
Main implementation / development work completed	
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	Enter a short summary of commit activity here. Add commit-history screenshots if applicable.
Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	
Next development steps	Outline the planned development tasks for the next two-week period.

Weeks 9-10	
Main implementation / development work completed	
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	Enter a short summary of commit activity here. Add commit-history screenshots if applicable.
Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	
Next development steps	Outline the planned development tasks for the next two-week period.

Weeks 11-12	
Main implementation / development work completed	
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	Enter a short summary of commit activity here. Add commit-history screenshots if applicable.
Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	
Next development steps	Outline the planned development tasks for the next two-week period.

Weeks 13-14	
Main implementation / development work completed	
Key files, folders, or components added / updated	
Evidence of versioning / commits / iterations	Enter a short summary of commit activity here. Add commit-history screenshots if applicable.
Testing, debugging, or validation carried out	
Problems encountered and how they were addressed	
Next development steps	Outline the planned development tasks for the next two-week period.